

Functional Requirement Document

Here are the detailed functional requirements based on the "Coding Requirements" section of the provided BRD:

FR-AURA-001: Parse AURA Input Payload

Title: Parse AURA Input Payload

Description: The system must decode, normalize, and transform AURA input payloads into a canonical XML string to drive subsequent field derivations and validations.

Preconditions:

- I_AURA_INPUT_BASE64 is not null.
- POLICY_NUMBER is not null.

Main Flow / Functional Steps:

1. Read I_AURA_INPUT_BASE64 and POLICY_NUMBER inputs.
2. Base64-decode I_AURA_INPUT_BASE64 and store in AURA_OUTPUT.
3. If AURA_OUTPUT starts with "{", attempt to parse as JSON and extract interviewDetails into AURA_OUTPUT.
4. Perform progressive decoding loop (max 100 attempts) until AURA_OUTPUT starts with "<".
5. If AURA_OUTPUT begins with "<", proceed; otherwise, log error.
6. Apply XSLT transformation based on Aura version (14 or 15) to produce canonical XML.
7. Output PARSED_AURA_OUTPUT and POLICY_NUMBER.

FR-FILE-001: Generate 10K File

Title: Generate 10K File

Description: The system must generate a 10K flat file with record-level fields derived through a maplet and file header/filename derivations.

Preconditions:

- Source data is available in T_APPLICATION_STAGING and T_BENEFICIARY.
- \$\$M_SRC_SQL is configured and valid.

Main Flow / Functional Steps:

1. Use \$\$M_SRC_SQL to select eligible applications and associated beneficiaries/policies.
2. Join application data with T_BENEFICIARY data using POLICY_NUMBER or application/beneficiary keys.
3. Derive Field_1..Field_n using mplt_BNCPLS_IF23_10K_Field_Calc maplet.
4. Derive File_Path_Name using EXP_Calc_10K_FileName expression.

5. Apply Transaction Control (TC_Seperate_10K_Files) to split output into multiple files (approx. 10,000 records per file).

6. Output FileName and Field_1..Field_n to File_Structure_10K flat file target.

FR-FILE-002: Split 10K File into Multiple Files

Title: Split 10K File into Multiple Files

Description: The system must split the output into multiple files, enforcing a maximum of approximately 10,000 records per physical file.

Preconditions:

Record count exceeds \$MAX_ROWS_PER_FILE (default: 10,000).

Main Flow / Functional Steps:

1. Use Transaction Control (TC_Seperate_10K_Files) to group rows into transactions.
2. Commit and initiate a new output file context when record count reaches \$MAX_ROWS_PER_FILE.
3. Derive unique filename for each output file using EXP_Calc_10K_FileName.

FR-MAP-001: Derive Field_1..Field_n using Mapplet

Title: Derive Field_1..Field_n using Mapplet

Description: The system must derive Field_1..Field_n using the mplt_BNCPLS_IF23_10K_Field_Calc mapplet.

Preconditions:

- Application and beneficiary/policy data is available from SQ_T_APPLICATION_STAGING.

Main Flow / Functional Steps:

1. Use mplt_BNCPLS_IF23_10K_Field_Calc to centralize business field derivations.
2. Apply field standardization, fixed-width/padded formatting, code translations, date normalization, and string cleansing as per 10K specification.

FR-EXP-001: Derive File_Path_Name using Expression

Title: Derive File_Path_Name using Expression

Description: The system must derive File_Path_Name using the EXP_Calc_10K_FileName expression.

Preconditions:

- Input data is available from upstream transformations.

Main Flow / Functional Steps:

1. Use EXP_Calc_10K_FileName to derive File_Path_Name based on interface code, business date/batch id, sequence, and extension.
2. Pass through Field_1..Field_n from mapplet to Transaction Control step.

FR-JAVA-001: Handle Errors and Logging in Java Transformation

Title: Handle Errors and Logging in Java Transformation

Description: The system must handle errors and logging in the Java transformation.

Preconditions:

- Exceptions occur during AURA payload parsing or XSLT transformation.

Main Flow / Functional Steps:

1. Log exception message and current AURA_OUTPUT string for diagnostic purposes.
2. Increment error count.
3. Proceed with row generation to allow partial processing.

FR-PARAM-001: Configure Parameters for Session/Workflow

Title: Configure Parameters for Session/Workflow

Description: The system must configure parameters for the session/workflow.

Preconditions:

- Parameters are defined in the parameter file or workflow variable.

Main Flow / Functional Steps:

1. Configure \$\$M_SRC_SQL, \$\$M_XSLT_FILE, \$\$AURA15_M_XSLT_FILE, \$\$OUT_DIR, \$\$FILE_PREFIX, \$\$FILE_EXT, \$\$BATCH_ID, \$\$RUN_ID, and \$\$MAX_ROWS_PER_FILE.
2. Ensure environment has read access to XSLT files and write permissions on \$\$OUT_DIR.